

16-811: Math Fundamentals for Robotics, Fall 2025

Assignment 3

DUE: Thursday, October 9, 2025, by 11:59pm

Regarding MATLAB/Python functions: It is possible that MATLAB/Python has some advanced functions that solve some of these problems for you directly from the data we have supplied. You may not use such functions. Instead, you should come up with your own solutions, based on the basic core functions available in MATLAB/Python. (When in doubt, ask.)

1. Consider the function $f(x) = \sin x - 0.5$ over the interval $[-\pi/2, \pi/2]$.
 - (a) What is the Taylor series expansion for $f(x)$ around $x = 0$?
 - (b) Graph $f(x)$ over the interval $[-\pi/2, \pi/2]$.
 - (c) Determine the best uniform approximation by a quadratic to the function $f(x)$ on the interval $[-\pi/2, \pi/2]$. What are the L_∞ and L_2 errors for this approximation?
 - (d) Determine the best least-squares approximation by a quadratic to the function $f(x)$ over the interval $[-\pi/2, \pi/2]$. What are the L_∞ and L_2 errors for this approximation?

Restriction: Do not use sampling-based methods.

Terminology: Suppose an approximation has error function $e(x)$, with x in interval $[a, b]$.

The L_∞ error is $\|e(x)\|_\infty = \max_{a \leq x \leq b} |e(x)|$ and the L_2 error is $\|e(x)\|_2 = \sqrt{\int_a^b |e(x)|^2 dx}$.

For your code submission, submit any code you used. In your pdf, please show all your hand derivations and code results, and replicate any code you would like the TAs to see.

2. Suppose very accurate values of some function $f(x)$ are given at the points $0 = x_0, x_1, \dots, x_{100} = 1$, with the $\{x_i\}$ uniformly distributed over the interval $[0, 1]$. (So $x_i = i/100$, $i = 0, \dots, 100$.)

The values $\{f(x_i)\}$ are given in the file ‘`problem2.txt`’ in sequential order (so, for example, $f(0.27) = f(x_{27}) = -0.964603914513021$).

Use the method of normal equations discussed in class, to find a description of $f(x)$ as the sum of a very few polynomials and cosines and sines. (You may be able to guess the answer since the function is fairly simple, but please also use the method of normal equations.)

For code, submit any code you used. In your pdf, replicate any code you would like the TAs to see, show your results and explain how you obtained them. If your search for a solution first considered some incorrect solutions, mention those and say how they informed your search for the correct solution.

[Hint: Try to express $f(x)$ as a linear combination of the functions, $1, x, x^2, \dots, \cos(\pi x), \sin(\pi x), \cos(2\pi x), \sin(2\pi x), \dots$. Use the method of normal equations. That method will not be enough by itself, since the underlying functions are redundant (more than a basis). However, that method is useful as a subroutine in a search. Try to find a very simple description of the function $f(x)$ by determining which coefficients in your sum may be set to zero. There may be multiple candidate answers; find one with the fewest nonzero coefficients. Graphing the function may be helpful in your search. You should need at most 3 nonzero coefficients when writing $f(x)$ as a sum of the functions $1, x, x^2, \dots, \cos(\pi x), \sin(\pi x), \cos(2\pi x), \sin(2\pi x), \dots$]

3. The *Chebyshev polynomials of the first kind*, $T_n(x)$, are defined indirectly on $[-1, 1]$ by:

$$T_n(\cos \theta) = \cos(n\theta), \text{ for } n \geq 0.$$

Expanding cosine, one finds the recurrence relation $T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x)$, for $n > 0$.

- (a) Derive T_4 and T_5 . (As always, please show your handwritten work.)
- (b) Without actually computing or working out any integrals, prove that $T_4(x)$ and $T_5(x)$ are orthogonal polynomials relative to the inner product

$$\langle g, h \rangle = \int_{-1}^1 (1 - x^2)^{-1/2} g(x) h(x) dx.$$

[Hint: Look carefully at the inner product and use a property of the polynomials.]

- (c) Recall that when we have an inner product on a vector space, we may define the *length of vector v* by $\|v\| = \sqrt{\langle v, v \rangle}$.

Here, we may view functions as vectors with an inner product defined by an integral as above. Then the length of $T_n(x)$ is the number $\sqrt{\langle T_n, T_n \rangle}$.

It turns out that all $T_n(x)$, with $n > 0$, have the same length.

Prove this fact by handcomputing the length of $T_n(x)$, that is, by working out the relevant integral (leave n symbolic, assume $n > 0$).

(Hint: You will likely find it useful to make the substitution $x = \cos \theta$ in the integral for $\langle T_n, T_n \rangle$. Don't forget to change the interval of integration as well.)

- (d) Finally, show that $\langle T_i, T_j \rangle = 0$ for all i and j such that $i \geq 0$, $j \geq 0$, and $i \neq j$.
(There are different ways to prove this, e.g., by working out an integral explicitly or by combining known facts from above and lecture.)

No code is expected or needed for any part of this problem.

In your pdf, please show all your derivations, proofs, and handwritten work.

[Another problem starts on the next page.]

4. After weeks of work you have finally completed construction of a gecko robot. It is a quadruped robot with suctioning feet that allow it to walk on walls. It is also equipped with a Kinect-like sensor, providing a 3D point cloud observation of the world. You want to use these point clouds to reason about the environment and aid in navigation.
- (a) You boot up the robot and place it on a table, taking an initial observation. The observation is saved in the provided `clear_table.txt`, and lists (x, y, z) locations in the following format:

$$\begin{array}{ccc} x_1 & y_1 & z_1 \\ & \vdots & \\ x_n & y_n & z_n \end{array}$$

Points are in units of meters and the positive x -direction is right, positive y -direction is down, and positive z -direction is forward. Find the least-squares approximation plane that fits the data. Visualize your fitted plane along with the data. What is the average distance of a point in the data set to the fitted plane?

Comment: The phrase “least-squares” is ambiguous. Below are descriptions of two possibilities that might occur to you. **Please use Linear Regression.** That approach is similar to our discussion of the normal equations in lecture.

Orthogonal-Distance Regression: In this approach, one computes the SVD decomposition of the $n \times 3$ matrix whose rows are the data points translated so their centroid is at the origin. It turns out that the third column of V is normal to a plane that minimizes the sum of squared orthogonal distances between the translated points and the plane. (This is a nice result to know.)

Linear Regression: This regression is based on the idea that, for perfectly planar data, all the points would satisfy a plane equation of the form $ax + by + cz + d = 0$. So one has a natural error $\sum_i (ax_i + by_i + cz_i + d)^2$, with i indexing the data points. One chooses the coefficients $\{a, b, c, d\}$ so as to minimize this error. In order to avoid degeneracies, one requires that not all of $\{a, b, c, d\}$ be 0.

Please use this approach.

(Additional comments: (i) One convenient approach is to set one of the coefficients $\{a, b, c, d\}$ to be 1 or -1 while letting the others vary in order to compute the best plane. (ii) Observe that $|ax_i + by_i + cz_i + d|$ is related to but not necessarily exactly the distance of the i^{th} data point from the plane described by $\{a, b, c, d\}$.)

- (b) Interested in your gecko robot, your cat jumps up on the table. You take a second observation, saved as the provided `cluttered_table.txt`. Using the same method as above, find the least-squares fit to the new data. How does it look? Why?
- (c) Can you suggest a way to still find a fit to the plane of the table regardless of clutter? Verify your idea by writing a program that can successfully find the dominant plane in a list of points regardless of outliers. [Hint: You may assume that the number of points in the plane is much larger than the number of points not in the plane.] Visualize `cluttered_table.txt` with your new plane.
- (d) Encouraged by your results when testing on a table, you move your geckobot into the hallway and take an observation saved as the provided `clean_hallway.txt`. Describe an extension to your solution to part (c) that finds the four dominant planes shown in the scene, then implement it and visualize the data and the four planes. You may assume that there are roughly the same amount of points in each plane.

- (e) You decide it is time to test your gecko robot's suction feet and move it to a different hallway. The feet are strong enough to ignore the force of gravity, allowing the robot to walk on the floor, walls, or ceiling. However, the locomotion of the legs works best on smooth surfaces with few obstacles. Using your solution from part (d), describe how you can mathematically characterize the smoothness of each surface. Load the provided scan `cluttered_hallway.txt`, find and plot the four wall planes, describe which surface is safest for your robot to traverse, and provide the smoothness scores from your mathematical characterization.

Note that you may *no longer* assume that there are roughly the same amount of points in each plane.

Parts (c), (d), and (e) intentionally leave room for some creativity and design. There may be several good approaches.

Please submit code for all parts (a), (b), (c), (d), (e) of this problem. In your pdf, explain what you did, how to run your code, and what results you obtained. Describe any design decisions you made. Include as well in your pdf any images you used to visualize data, and explain their meanings. Finally, replicate in your pdf any code you would like the TAs to see.

1. Consider the function $f(x) = \sin x - 0.5$ over the interval $[-\pi/2, \pi/2]$.

- What is the Taylor series expansion for $f(x)$ around $x = 0$?
- Graph $f(x)$ over the interval $[-\pi/2, \pi/2]$.
- Determine the best uniform approximation by a quadratic to the function $f(x)$ on the interval $[-\pi/2, \pi/2]$. What are the L_∞ and L_2 errors for this approximation?
- Determine the best least-squares approximation by a quadratic to the function $f(x)$ over the interval $[-\pi/2, \pi/2]$. What are the L_∞ and L_2 errors for this approximation?

Restriction: Do not use sampling-based methods.

Terminology: Suppose an approximation has error function $e(x)$, with x in interval $[a, b]$.

The L_∞ error is $\|e(x)\|_\infty = \max_{a \leq x \leq b} |e(x)|$ and the L_2 error is $\|e(x)\|_2 = \sqrt{\int_a^b |e(x)|^2 dx}$.

For your code submission, submit any code you used. In your pdf, please show all your hand derivations and code results, and replicate any code you would like the TAs to see.

$$f(x) = \sin x - 0.5 \quad \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

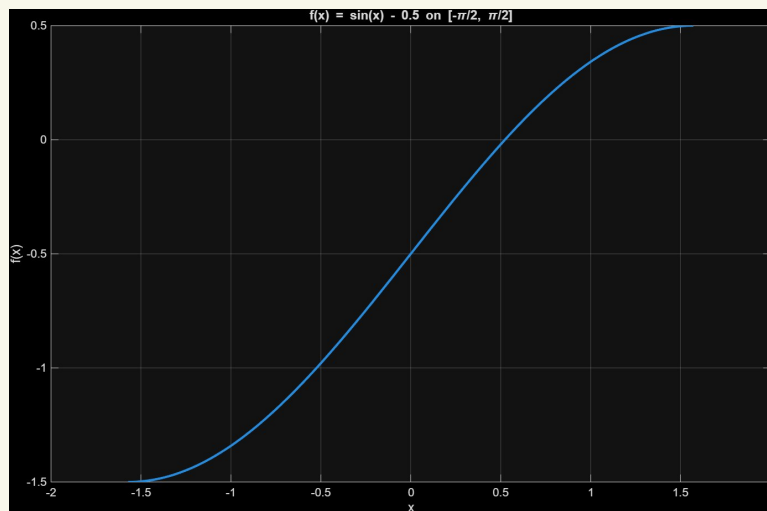
$$a). \quad f' = \cos x \quad f'' = -\sin x \quad f''' = -\cos x$$

$$f(x) = f(0) + \frac{f'(0)}{1} x + \frac{f''(0)}{2!} x^2 + \frac{f'''(0)}{3!} x^3 + \dots$$

$$= -0.5 + x - \frac{1}{6} x^3 + \frac{1}{120} x^5 + \dots$$

$$f(x) = -0.5 + \sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n+1)!} x^{2n+1}$$

b).



c). $P(x) = a + bx + cx^2$

$$E = f(x) - p(x) = \sinh x - 0.5 - a - bx - cx^2$$

$$\begin{cases} \sin x_1 - 0.5 - a - bx_1 - cx_1^2 = E \\ \sin x_2 - 0.5 - a - bx_2 - cx_2^2 = -E \\ x_3 = B \\ x_4 = -B \end{cases}$$

On the symmetric interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$, $f(x)$ is nearly odd while x^2 is even. the quadratic term does not help reduce the uniform error.

$$\Rightarrow -1.5 - a + \frac{7}{2}b = 7$$

$$-a - 0.5 + 9hX_2 - 6X_2 = -B.$$

$$-9 - 0.5 + 51x_2 - 6x_3 = E$$

$$0.5 - a - \frac{3}{2}b = -E.$$

$$e\left(\frac{z}{2}\right) + e\left(-\frac{z}{2}\right) = 0 \Rightarrow (-1.5 - a + \frac{3}{2}b) + (0.5 - a - \frac{3}{2}b) = 0.$$

$$a = -0.5$$

$$\Rightarrow -1 + \frac{3}{2}b = E$$

$$\sin x_v - bx_v = -E$$

$$\sin x_3 - bx_3 = R$$

$$1 - \frac{3}{2}b = -\frac{1}{2}$$

$$-1 + \frac{7}{2}b + 91h x_2 - bx_2 = 0$$

$$b = \frac{\sinh x_2 - 1}{x_2 - \pi/2}$$

$$1 - \frac{x}{2}b + \sinh x_3 - b x_3 = 0$$

$$e'(x_2) = e'(x_3) = 0 = \cosh x - b.$$

$$\cosh x_3 - \frac{\sinh x_3 + 1}{x_3 + \frac{x}{2}} = 0.$$

$$e'(x_2) = \cosh x_1 - b$$

$$e'(x_3) = \cosh x_3 - b.$$

$$\sinh x_3 + 1 - x_3 \cosh x_3 - \frac{x}{2} \cosh x_1 = 0 \Rightarrow x_2 = -x_3$$

$$\text{Muller's function: } x = 0.76033 \quad b = 0.7246 \quad a = 0.5$$

$$\|f - p\|_{\infty} = \|e(\frac{x}{2})\| = \|1 - \frac{x}{2} \cdot 0.7246\| = 0.1382$$

$$\|f - p\|_2 = \sqrt{\int_{-\frac{x}{2}}^{\frac{x}{2}} (\cosh x - bx)^2 dx} = 0.1704.$$

$$d) \cdot \int_{-h}^h (f(x) - p(x)) \phi_i(x) dx = 0. \quad \phi_i \in \{1, x, x^2\}$$

$$\text{for } \phi = x$$

$$\int_{-h}^h (0.5 + \sinh x - a - bx - cx^2) x dx = 0$$

$$\int_{-h}^h \cancel{\frac{1}{2}x} + x \sinh x - \cancel{ax} - bx^2 - \cancel{cx^3} dx = 0$$

$$\int_{-h}^h x \sinh x dx - \int_{-h}^h bx^2 dx = 0$$

$$b = \frac{2(-h \cosh h + \sinh h)}{2h^3/3} \quad (h = \frac{x}{2})$$

$$= \frac{24}{x^3}$$

$$\phi = 1 \quad \int_{-h}^h c \sin x + 0.5 - a - bx - cx^2 dx = 0$$

$$\int_{-h}^h a \sin x = 0 \quad \int_{-h}^h bx = 0 \Rightarrow h - 2ah - \frac{2}{3}ch^3 = 0$$

$$2a + \frac{2}{3}ch^2 = 1$$

$$\phi = x^2 \quad \int_{-h}^h x^2 (c \sin x + \frac{1}{2} - a - bx - cx^2) dx = 0$$

$$\int_{-h}^h x^2 \sin x \quad \& \quad \int_{-h}^h x^3 = 0$$

$$(\frac{1}{2} - a) \int_{-h}^h x^2 dx - c \int_{-h}^h x^4 dx = 0$$

$$(\frac{1}{2} - a) \frac{2h^3}{3} - c \frac{2h^5}{5} = 0$$

$$5h^3 - 10h^3a - 3ch^5 = 0$$

$$\begin{cases} 10a + 3ch^2 = 5 \\ 2a + \frac{2}{3}ch^2 = 1 \end{cases} \Rightarrow \begin{matrix} a = 0.5 \\ c = 0 \end{matrix}$$

$$\text{we have } p(x) = 0.5 + \frac{24}{25}x = 0.5 + 0.774x$$

$$L_0 = |e(\frac{\pi}{2})| = 0.21585$$

$$L_2 = \sqrt{\int_{-\pi/2}^{\pi/2} (c \sin x - 0.774)^2 dx} = 0.1507$$

3. The Chebyshev polynomials of the first kind, $T_n(x)$, are defined indirectly on $[-1, 1]$ by:

$$T_n(\cos \theta) = \cos(n\theta), \text{ for } n \geq 0.$$

Expanding cosine, one finds the recurrence relation $T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x)$, for $n > 0$.

- (a) Derive T_4 and T_5 . (As always, please show your handwritten work.)
- (b) Without actually computing or working out any integrals, prove that $T_4(x)$ and $T_5(x)$ are orthogonal polynomials relative to the inner product

$$\langle g, h \rangle = \int_{-1}^1 (1-x^2)^{-1/2} g(x)h(x)dx.$$

[Hint: Look carefully at the inner product and use a property of the polynomials.]

- (c) Recall that when we have an inner product on a vector space, we may define the *length* of vector v by $\|v\| = \sqrt{\langle v, v \rangle}$.

Here, we may view functions as vectors with an inner product defined by an integral as above. Then the length of $T_n(x)$ is the number $\sqrt{\langle T_n, T_n \rangle}$.

It turns out that all $T_n(x)$, with $n > 0$, have the same length.

Prove this fact by handcomputing the length of $T_n(x)$, that is, by working out the relevant integral (leave n symbolic, assume $n > 0$).

(Hint: You will likely find it useful to make the substitution $x = \cos \theta$ in the integral for $\langle T_n, T_n \rangle$. Don't forget to change the interval of integration as well.)

- (d) Finally, show that $\langle T_i, T_j \rangle = 0$ for all i and j such that $i \geq 0$, $j \geq 0$, and $i \neq j$. (There are different ways to prove this, e.g., by working out an integral explicitly or by combining known facts from above and lecture.)

No code is expected or needed for any part of this problem.

In your pdf, please show all your derivations, proofs, and handwritten work.

a). $T_0(x) = 1$

$$T_1(x) = x \quad T_2(x) = 2xT_1(x) - T_0(x) = 2x^2 - 1$$

$$T_3(x) = 2xT_2(x) - T_1(x) = 4x^3 - 3x$$

$$T_4(x) = 2xT_3(x) - T_2(x) = 2x(4x^3 - 3x) - (2x^2 - 1) \\ = 8x^4 - 8x^2 - 1$$

$$T_5(x) = 2xT_4(x) - T_3(x) = 2x(8x^4 - 8x^2 - 1) - (4x^3 - 3x) \\ = 16x^5 - 20x^3 + 5x$$

b). $\langle T_4(x), T_5(x) \rangle = \int_{-1}^1 (1-x^2)^{-1/2} (8x^4 - 8x^2 - 1)(16x^5 - 20x^3 + 5x) dx.$

let $x = \cos \theta \Rightarrow \theta \in [0, \pi]$ $dx = -\sin \theta d\theta$
 $\sqrt{1-x^2} = \sin \theta.$

$$\frac{dx}{\sqrt{1-x^2}} = \frac{-\sin \theta d\theta}{\sin \theta} = -d\theta.$$

$$\int_0^{\pi} T_4(\cos\theta) T_5(\cos\theta) d\theta$$

$$\Rightarrow \int_0^{\pi} \cos(4\theta) \cos(5\theta) d\theta = 0 \Rightarrow \text{we have two wave orthogonal at } [0, \pi]$$

$$\cos(4\theta) \cos(5\theta) = \frac{1}{2} \int_0^{\pi} \cos(-\theta) d\theta + \frac{1}{2} \int_0^{\pi} \cos(9\theta) d\theta$$

$$= 0.$$

$\therefore \langle T_4, T_5 \rangle = 0$ orthogonal polynomials relative to inner product so any $\langle T_n, T_m \rangle = 0$.

c). let $x = \cos\theta$ we have $\theta \in [0, \pi]$.

$$dx = -\sin\theta d\theta \quad \sqrt{1-x^2} = \sin\theta \quad \frac{dx}{\sqrt{1-x^2}} = -d\theta.$$

$$\langle T_n, T_n \rangle = \int_{-1}^1 -T_n(\cos\theta)^2 d\theta = \int_0^{\pi} \cos^2(n\theta) d\theta.$$

$$\cos^2 u = \frac{1}{2} (1 + \cos 2u) = \frac{1}{2} \int_0^{\pi} (1 + \cos(2n\theta)) d\theta = \frac{1}{2} \pi$$

$$\|T_n\| = \sqrt{\frac{\pi}{2}}.$$

$\Rightarrow$ there is no n -term in the final form $\Rightarrow$ for any $n > 0$, has the same length.

d). $x = \cos\theta$.

$$\langle T_i, T_j \rangle = \int_0^{\pi} \cos(i\theta) \cos(j\theta) d\theta$$

$$= \frac{1}{2} \int_0^{\pi} \cos(i-j)\theta d\theta + \frac{1}{2} \int_0^{\pi} \cos(i+j)\theta d\theta$$

$$= 0$$

$$\int_0^{\pi} \cos k\theta d\theta = \left[\frac{1}{k} \sin k\theta \right]_0^{\pi} = 0 \Rightarrow \text{when } i \neq j \quad i+j \neq 0$$

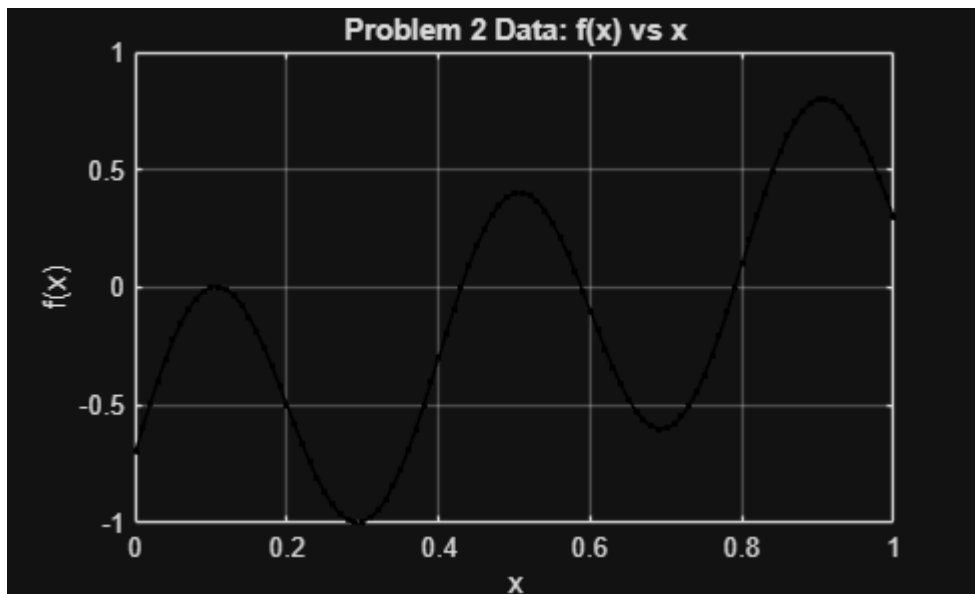
$$\langle T_i, T_j \rangle = 0$$

HW3 Question 2

```
clear; clc; close all;

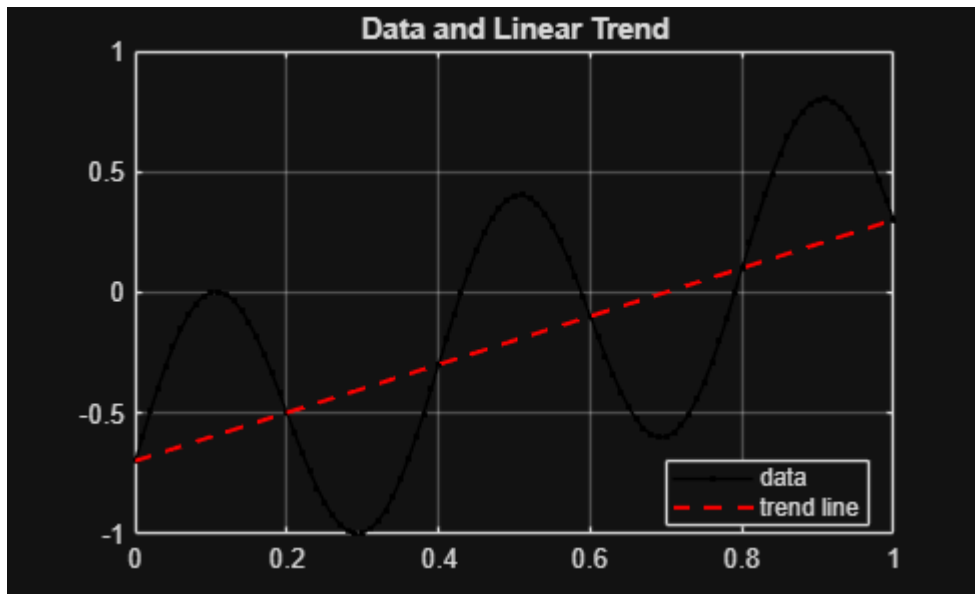
x = 0:0.01:1;
y = load("problem2.txt");
y = y(:);
N = numel(y);
assert(numel(x) == N, 'x and y must have the same length (101).');

figure(1);
plot(x, y, 'k.-');
xlabel('x'); ylabel('f(x)');
title('Problem 2 Data: f(x) vs x');
grid on;
```

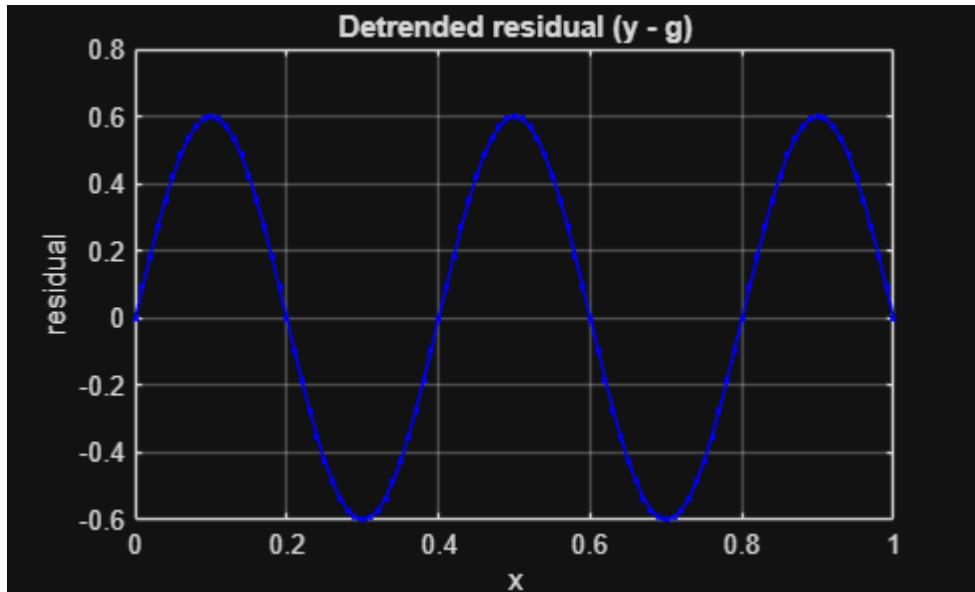


```
y0 = y(1);
y1 = y(end);
g = (y1 - y0) * x + y0;

figure(2);
plot(x, y, 'k.-'); hold on;
plot(x, g, 'r--', 'LineWidth', 1.5);
legend('data', 'trend line', 'Location', 'best');
title('Data and Linear Trend');
grid on;
```



```
t = y - g.';
figure(3);
plot(x, t, 'b.-');
xlabel('x'); ylabel('residual');
title('Detrended residual (y - g)');
grid on;
```



```
M = [x.' sin(6*pi*x).'];

A = (M.' * M) \ (M.' * y);

a1 = A(1);
```

```
a2 = A(2);
fprintf('Recovered model:\n');
```

Recovered model:

```
fprintf(' f(x) ≈ %.6f * x + %.6f * sin(6*pi*x)\n', a1, a2);
```

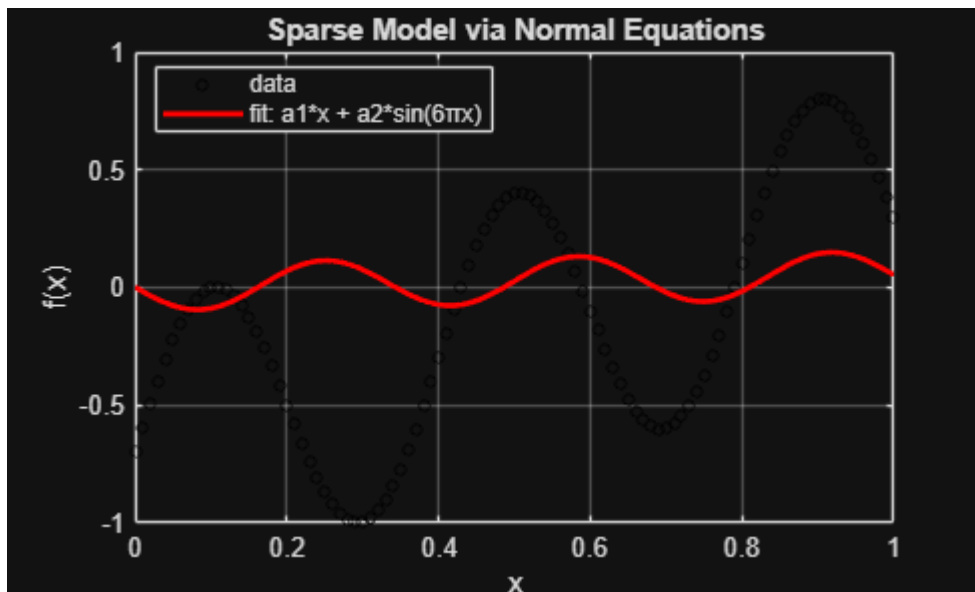
$f(x) \approx 0.052210 * x + -0.100266 * \sin(6\pi x)$

```
yhat = M * A;
err2 = norm(y - yhat, 2);
```

```
fprintf(' Residual 2-norm ||y - yhat||_2 = %.3e\n\n', err2);
```

Residual 2-norm $\|y - \hat{y}\|_2 = 5.189e+00$

```
figure(4);
plot(x, y, 'ko', 'MarkerSize', 4, 'DisplayName', 'data'); hold on;
plot(x, yhat, 'r-', 'LineWidth', 2, 'DisplayName', 'fit: a1*x + a2*sin(6πx)');
xlabel('x'); ylabel('f(x)');
title('Sparse Model via Normal Equations');
legend('Location', 'best');
grid on;
```

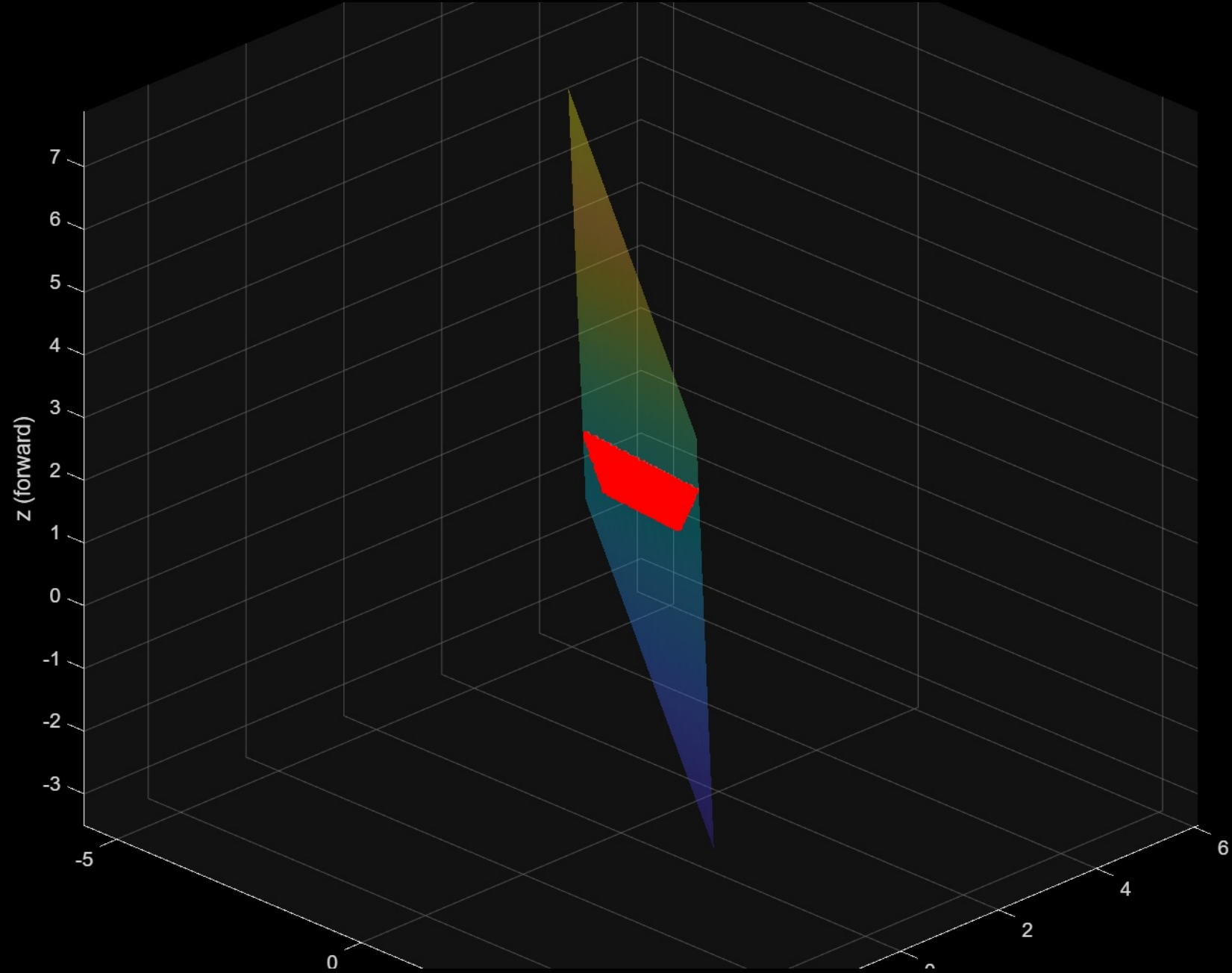


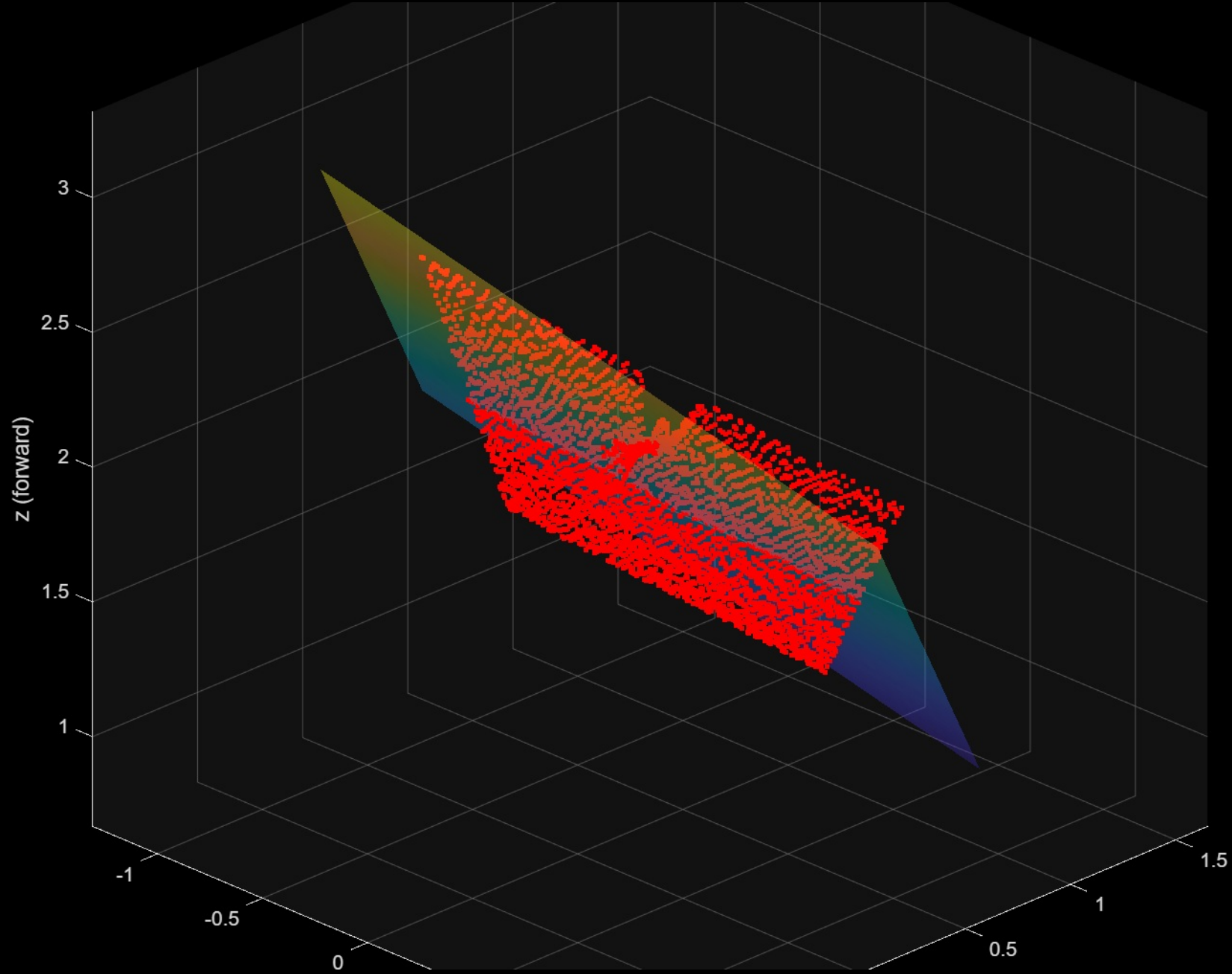
```

1 function HW3_4ab()
2
3     clc; close all;
4
5     data_file1 = 'clear_table.txt';
6     fprintf('==== Part (a): %s ==== \n', data_file1);
7     [t1, rss1, avg1] = q4_a_b(data_file1, true);
8     fprintf('Plane (a): z = %.6f x + %.6f y + %.6f \n', t1(1), t1(2), t1(3));
9     fprintf('RSS^0.5 (z residual): %.6f \n', rss1);
10    fprintf('Mean point-to-plane distance: %.6f m \n \n', avg1);
11
12    data_file2 = 'cluttered_table.txt';
13    fprintf('==== Part (b): %s ==== \n', data_file2);
14    [t2, rss2, avg2] = q4_a_b(data_file2, true);
15    fprintf('Plane (b): z = %.6f x + %.6f y + %.6f \n', t2(1), t2(2), t2(3));
16    fprintf('RSS^0.5 (z residual): %.6f \n', rss2);
17    fprintf('Mean point-to-plane distance: %.6f m \n', avg2);
18 end
19
20
21 function [t, rss, avg_dist] = q4_a_b(data_file, do_plot)
22
23
24    P = load(data_file);
25    if size(P,2) ~= 3
26        error('Expected an N x 3 text file with columns [x y z].');
27    end
28    x = P(:,1); y = P(:,2); z = P(:,3);
29
30    A = [x, y, ones(length(x),1)];
31    t = A \ z;
32
33    zhat = A * t;
34    rss = sqrt(sum((z - zhat).^2));
35
36    a = t(1); b = t(2); c = t(3);
37    numer = abs(a.*x + b.*y - z + c);
38    denom = sqrt(a^2 + b^2 + 1);
39    avg_dist = mean(numer ./ denom);
40
41    if do_plot
42        plot_points_and_plane(x, y, z, t, sprintf('%s - plane fit', data_file));
43    end
44 end
45
46
47 function plot_points_and_plane(x, y, z, t, ttl)
48
49    a = t(1); b = t(2); c = t(3);
50
51    pad = 0.05;
52    xs = linspace(min(x)-pad, max(x)+pad, 40);
53    ys = linspace(min(y)-pad, max(y)+pad, 40);
54    [X, Y] = meshgrid(xs, ys);
55    Z = a.*X + b.*Y + c;
56
57    figure; hold on; grid on; axis vis3d;
58    scatter3(x, y, z, 8, 'r', 'filled');
59    surf(X, Y, Z, 'FaceAlpha', 0.35, 'EdgeColor', 'none');
60
61    xlabel('x (right)'); ylabel('y (down)'); zlabel('z (forward)');

```

```
62     title(sprintf('%s\nz = %.4f x + %.4f y + %.4f', ttl, a, b, c));
63     view(45, 25);
64 end
65
```





```

1 function HW3_q4_cd()
2     clc; close all;
3
4     % ===== Part (c): dominant plane via RANSAC =====
5     file_c = 'cluttered_table.txt'; % adjust as needed
6     fprintf('==== Part (c): %s =====\n', file_c);
7     [pl_c, inliers_c, stats_c] = q4_c(file_c, 0.01, 4000);
8     fprintf('RANSAC plane (normalized): a=%.6f, b=%.6f, c=%.6f, d=%.6f\n', ...
9         pl_c.a, pl_c.b, pl_c.c, pl_c.d);
10    fprintf('Inliers: %d / %d (%.1f%%)\n', stats_c.inliers, stats_c.N, 100*stats_c.inlier_ratio);
11    fprintf('Mean |distance| on inliers: %.6f m\n\n', stats_c.avg_abs_dist);
12
13    % ===== Part (d): four dominant planes =====
14    file_d = 'clean_hallway.txt'; % adjust as needed
15    fprintf('==== Part (d): %s =====\n', file_d);
16    [planes_d, masks_d] = q4_d(file_d, 0.01, 4000, 200);
17    fprintf('Found %d plane(s)\n', numel(planes_d));
18    for i = 1:numel(planes_d)
19        pl = planes_d{i};
20        fprintf(' Plane %d: a=%.6f, b=%.6f, c=%.6f, d=%.6f\n', i, pl.a, pl.b, pl.c, pl.d);
21    end
22 end
23
24 % =====
25 function [plane, inlier_mask, stats] = q4_c(data_file, thresh, iters)
26 % Part (c): RANSAC dominant plane on a cluttered table
27 % Returns:
28 %   plane: struct with fields a,b,c,d (||[a b c]|| = 1, d signed)
29 %   inlier_mask: Nx1 logical for all points in data file
30 %   stats: struct with counts and mean inlier distance
31
32 P = load_xyz(data_file);
33 [plane, inlier_mask] = ransac_plane(P, thresh, iters);
34 Pin = P(inlier_mask, :);
35 avg_abs = mean(point_plane_abs_distance(Pin, plane));
36
37 stats = struct();
38 stats.N = size(P,1);
39 stats.inliers = nnz(inlier_mask);
40 stats.inlier_ratio = stats.inliers / stats.N;
41 stats.avg_abs_dist = avg_abs;
42
43 % Plot
44 ttl = sprintf('Part (c): %s - dominant plane via RANSAC', data_file);
45 plot_with_planes(P, {plane}, {inlier_mask}, ttl);
46 end
47
48 % =====
49 function [planes, masks] = q4_d(data_file, thresh, iters, minCluster)
50 % Part (d): Extract four dominant planes iteratively (peel-off)
51 % Returns:
52 %   planes: 1xK cell of plane structs
53 %   masks : 1xK cell of Nx1 logical masks for original indexing
54
55 if nargin < 4 || isempty(minCluster), minCluster = 200; end
56 P = load_xyz(data_file);
57 [planes, masks] = extract_k_planes(P, 4, thresh, iters, minCluster);
58 ttl = sprintf('Part (d): %s - four dominant planes', data_file);
59 plot_with_planes(P, planes, masks, ttl);
60 end
61

```

```

62 % ===== RANSAC Core =====
63 function [best_plane, best_mask] = ransac_plane(P, thresh, iters)
64 % RANSAC to find a dominant plane; refine with LS on inliers
65
66     N = size(P,1);
67     best_cnt = -1;
68     best_plane = [];
69     best_mask = false(N,1);
70
71     for t = 1:iters
72         idx = randperm(N,3);
73         pl_cand = plane_from_3(P(idx(1),:), P(idx(2),:), P(idx(3),:));
74         if isempty(pl_cand), continue; end
75
76         d = point_plane_abs_distance(P, pl_cand);
77         mask = d <= thresh;
78         cnt = nnz(mask);
79         if cnt > best_cnt
80             best_cnt = cnt;
81             best_plane = pl_cand;
82             best_mask = mask;
83         end
84     end
85
86     if isempty(best_plane), error('RANSAC failed to find a plane.');
```

end

```

88     % Refine by LS on inliers, keeping the representation consistent
89     Pin = P(best_mask, :);
90     pl_ref = fit_plane_ls_fixed_d(Pin, -1); %  $z = a x + b y + c \Rightarrow ax + by - z + c = 0$ 
91     best_plane = pl_ref;
92 end
93
94 function [planes, masks] = extract_k_planes(P, k, thresh, iters, minCluster)
95 % Iteratively peel off k dominant planes
96
97     Rem = P;
98     idx_map = (1:size(P,1)).';
99     planes = {};
100     masks = {};
101
102     for i = 1:k
103         if size(Rem,1) < 3, break; end
104         try
105             [pl, mask_rem] = ransac_plane(Rem, thresh, iters);
106         catch
107             break
108         end
109         if nnz(mask_rem) < minCluster
110             break
111         end
112         planes{end+1} = pl; %#ok<AGROW>
113
114         % Map inliers back to original indexing
115         in_global = false(size(P,1),1);
116         in_global(idx_map(mask_rem)) = true;
117         masks{end+1} = in_global; %#ok<AGROW>
118
119         % Peel off
120         Rem = Rem(~mask_rem, :);
121         idx_map = idx_map(~mask_rem);
122     end

```

```

123 end
124
125 % ===== Geometry Utils =====
126 function P = load_xyz(path)
127     P = load(path);
128     if size(P,2) ~= 3
129         error('Expected an N x 3 text file with columns [x y z].');
130     end
131 end
132
133 function pl = fit_plane_ls_fixed_d(P, d_fixed)
134 % Fit  $ax + by + cz + d = 0$  with fixed d (use c = -1 to match  $z = a x + b y + c$ )
135 % Here we choose c = -1 implicitly by fitting  $z = a x + b y + c$  first
136
137     % Solve  $z = a*x + b*y + c$  via LS
138     x = P(:,1); y = P(:,2); z = P(:,3);
139     A = [x, y, ones(size(x))];
140     t = A \ z; % t = [a; b; c]
141     a = t(1); b = t(2); c0 = t(3);
142
143     % Convert to  $ax + by + cz + d = 0$  form with c = -1, d = c0
144     pl = normalize_plane(struct('a', a, 'b', b, 'c', -1, 'd', c0));
145 end
146
147 function pl = plane_from_3(p1, p2, p3)
148 % Plane through three points; normalized so  $||[a\ b\ c]|| = 1$ 
149
150     v1 = p2 - p1; v2 = p3 - p1;
151     n = cross(v1, v2);
152     nrm = norm(n);
153     if nrm < 1e-12
154         pl = [];
155         return;
156     end
157     n = n / nrm;
158     d = -dot(n, p1);
159     pl = normalize_plane(struct('a', n(1), 'b', n(2), 'c', n(3), 'd', d));
160 end
161
162 function pl = normalize_plane(pl)
163 % Normalize so  $|(a,b,c)| = 1$  and flip sign so  $d \leq 0$  for consistency
164
165     nrm = norm([pl.a pl.b pl.c]);
166     if nrm == 0, return; end
167     pl.a = pl.a/nrm; pl.b = pl.b/nrm; pl.c = pl.c/nrm; pl.d = pl.d/nrm;
168     if pl.d > 0
169         pl.a = -pl.a; pl.b = -pl.b; pl.c = -pl.c; pl.d = -pl.d;
170     end
171 end
172
173 function d = point_plane_abs_distance(P, pl)
174 %  $|ax + by + cz + d| / ||n||$ 
175     num = abs(P * [pl.a; pl.b; pl.c] + pl.d);
176     den = sqrt(pl.a^2 + pl.b^2 + pl.c^2);
177     d = num / den;
178 end
179
180 % ===== Visualization =====
181 function plot_with_planes(P, planes, masks, ttl)
182 % P: Nx3, planes: {plane1, plane2, ...}, masks: {mask1, ...} or {}
183     if nargin < 3, masks = {}; end

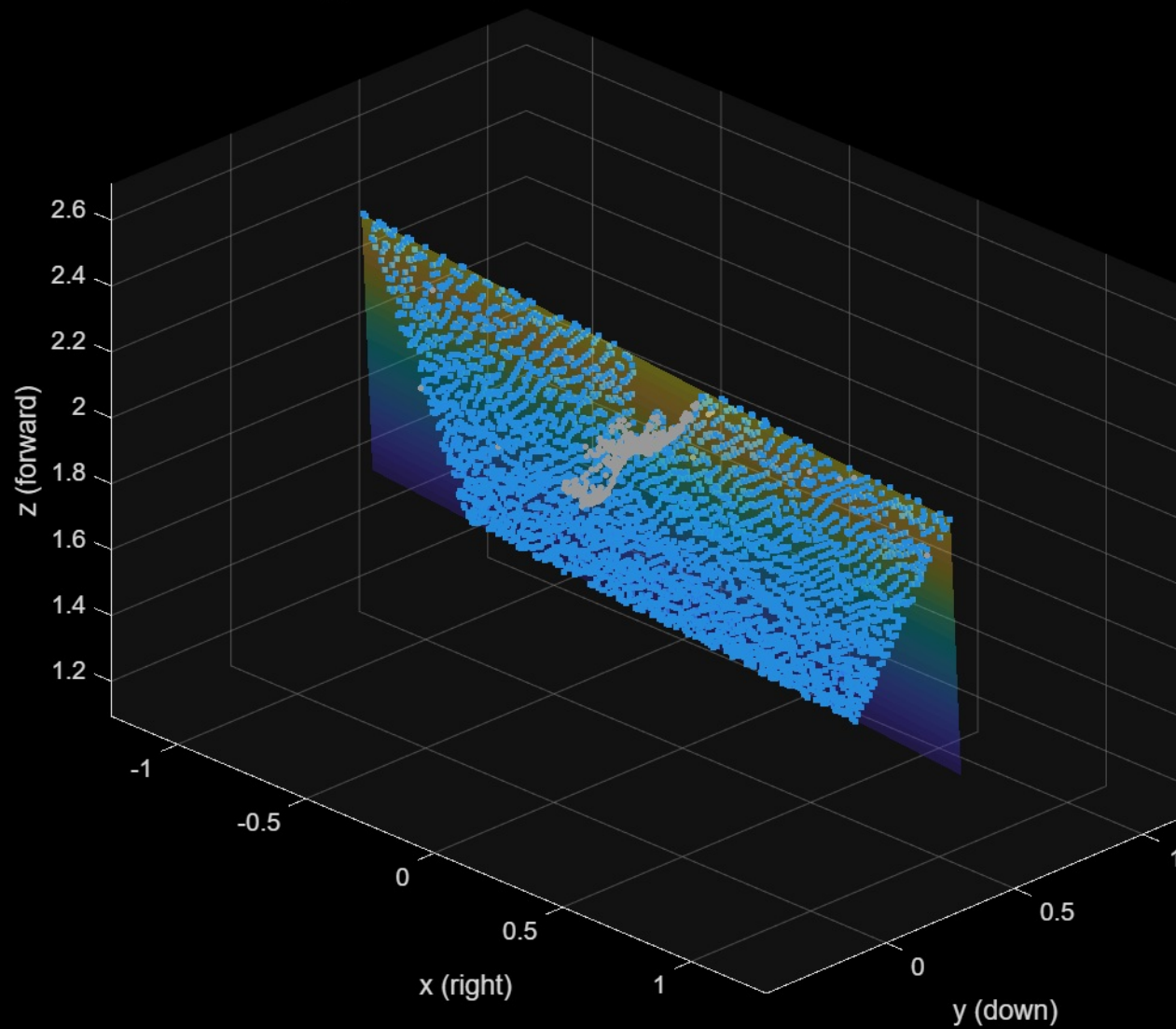
```

```

184
185 figure; hold on; grid on; axis equal; view(45,25);
186
187 plotted = false(size(P,1),1);
188 if ~isempty(masks)
189     C = lines(max(1, numel(masks)));
190     for i = 1:numel(masks)
191         m = masks{i};
192         scatter3(P(m,1), P(m,2), P(m,3), 8, C(i,:), 'filled');
193         plotted = plotted | m;
194     end
195 end
196 if any(~plotted)
197     scatter3(P(~plotted,1), P(~plotted,2), P(~plotted,3), 6, [0.6 0.6 0.6], 'filled');
198 end
199
200 % Extents for plane patches
201 x_min = min(P(:,1)); x_max = max(P(:,1));
202 y_min = min(P(:,2)); y_max = max(P(:,2));
203 z_min = min(P(:,3)); z_max = max(P(:,3));
204 xs = linspace(x_min, x_max, 28);
205 ys = linspace(y_min, y_max, 28);
206 zs = linspace(z_min, z_max, 28);
207
208 for i = 1:numel(planes)
209     pl = planes{i};
210     a = pl.a; b = pl.b; c = pl.c; d = pl.d;
211     if abs(c) >= max(abs(a),abs(b))
212         [X,Y] = meshgrid(xs, ys);
213         Z = (-(a*X + b*Y + d)) / c;
214     elseif abs(b) >= max(abs(a),abs(c))
215         [X,Z] = meshgrid(xs, zs);
216         Y = (-(a*X + c*Z + d)) / b;
217     else
218         [Y,Z] = meshgrid(ys, zs);
219         X = (-(b*Y + c*Z + d)) / a;
220     end
221     surf(X, Y, Z, 'FaceAlpha', 0.35, 'EdgeColor', 'none');
222 end
223
224 xlabel('x (right)'); ylabel('y (down)'); zlabel('z (forward)');
225 title(ttl);
226 end
227

```

Part (c): cluttered_table.txt — dominant plane via RANSAC



Part (d): clean_nallway.txt — four dominant planes

